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Sri Adichunchanagiri Shikshana Trust (R.)

SJC INSTITUTE OF TECHNOLOGY

(An Autonomous Institution under VTU from 2024-25)

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Chikkaballapura - 562 101, Karnataka



Estd. 1986

Department of Civil Engineering

Tutorial - II

Subject: WATER CONSERVATION AND RAIN WATER HARVESTING – BCV654A

1. Explain the feasibility of implementing rooftop rainwater harvesting in rural households?
2. Discuss the major sources of water pollution in rural and urban areas?
3. Explain the advantages of contour trenching towards water conservation in hilly regions?
4. Describe the environmental and economic benefits of rainwater harvesting?
5. Explain the effect of soil type and permeability on groundwater recharge rates?
6. How the traditional knowledge be integrated with modern technology for sustainable water management?
7. A community building with a roof area of 300 m^2 receives an average annual rainfall of 600 mm. The runoff coefficient is 0.9. Calculate the total annual rainwater harvested.
8. Explain the procedure to determine the size and capacity of storage tanks for rainwater harvesting?
9. How the traditional systems like stepwells, tanks, and johads help in water conservation?
10. Design a rainwater harvesting system for a small residential house with the following details: Roof Area: 150 m^2 , Average Annual Rainfall: 1000 mm Runoff Coefficient: 0.8 Storage Tank Capacity: The tank should store rainwater to meet the household's needs for 3 months. The daily water requirement for the household is 500 liters. Design total volume of rainwater harvested per year, Size of the storage tank, Overflow and filtration system.
11. Explain the importance of ground water recharge. What are the factors affecting ground water recharge?
12. Differentiate between rooftop rainwater harvesting and farm ponding
13. Differentiate between modern and traditional water harvesting techniques.
14. Design a comprehensive water harvesting plan for a drought-prone rural village.
15. Design a rainwater harvesting structure suitable for a semi-urban area.